3.5.5	**Reactions of Inorganic Compounds in Aqueous Solution**	
	Lewis acids and bases	know the definitions of a Lewis acid and Lewis base; understand the importance of lone pair electrons in co-ordinate bond formation
	Metal-aqua ions	know that metal–aqua ions are formed in aqueous solution: $[M(H_2O)_6]^{2+}$, limited to M = Fe, Co and Cu $[M(H_2O)_6]^{3+}$, limited to M = Al, Cr and Fe
	Acidity or hydrolysis reactions	understand the equilibria $[M(H_2O)_6]^{2+} + H_2O \rightleftharpoons M(H_2O)_5(OH)]^{+} + H_3O^{+}$ and $[M(H_2O)_6]^{3+} + H_2O \rightleftharpoons [M(H_2O)_5(OH)]^{2+} + H_3O^{+}$ to show generation of acidic solutions with M^{3+}, and very weakly acidic solutions with M^{2+}

		understand that the acidity of $[M(H_2O)_6]^{3+}$ is greater than that of $[M(H_2O)_6]^{2+}$ in terms of the (charge/size ratio) of the metal ion
		be able to describe and explain the simple test-tube reactions of M^{2+} (aq) ions, limited to M = Fe, Co and Cu, and of M^{3+} (aq) ions, limited to M = Al, Cr and Fe, with the bases OH^-, NH_3 and CO_3^{2-}
		know that MCO_3 is formed but that $M_2(CO_3)_3$ is not formed
		know that some metal hydroxides show amphoteric character by dissolving in both acids and bases (e.g. hydroxides of Al^{3+} and Cr^{3+})
		know the equilibrium reaction $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$
	Substitution reactions	understand that the ligands NH_3 and H_2O are similar in size and are uncharged, and that ligand exchange occurs without change of co-ordination number (e.g. Co^{2+} and Cr^{3+})
		know that substitution may be incomplete (e.g. the formation of $[Cu(NH_3)_4(H_2O)_2]^{2+}$)
		understand that the Cl^- ligand is larger than these uncharged ligands and that ligand exchange can involve a change of co-ordination number (e.g. Co^{2+} and Cu^{2+})
		know that substitution of unidentate ligand with a bidentate or a multidentate ligand leads to a more stable complex
		understand this chelate effect in terms of a positive entropy change in these reactions